

### 1. Finding Mean Using Python

```
In [1]: data = [10,20,30,40,50]
mean_val = sum(data)/len(data)
print("Mean:",mean_val)
```

Mean: 30.0

### 2. Finding Median Using Python

```
In [2]: data = [40,10,30,50,20]
data.sort()
n = len(data)
if n%2 == 0:
    median_val = (data[n//2-1] + data[n//2]) / 2
else:
    median_val = data[n//2]
print("Median:",median_val)
```

Median: 30

### 3. Finding Mode Using Python

```
In [5]: from collections import Counter

data = [1,2,2,3,4,2,5]
count = Counter(data)
mode_val = count.most_common(1)[0][0]
print("Mode:",mode_val)
```

Mode: 2

### 4. Finding Range Using Python

```
In [6]: data = [15,8,22,42,10]

range_val = max(data) - min(data)
print("Range:",range_val)
```

Range: 34

### 5. Finding Variance Using Python (Population)

```
In [7]: data = [2,4,4,4,5,5,7,9]
mean_val = sum(data)/len(data)
variance_val = sum((x - mean_val) ** 2 for x in data) / len(data)
print("Variance:",variance_val)
```

Variance: 4.0

### 6. Finding Standard Deviation Pure Python

```
In [8]: data = [2,4,4,4,5,5,7,9]
mean_val = sum(data)/len(data)
variance_val = sum((x - mean_val) ** 2 for x in data) / len(data)
std_dev = variance_val ** 0.5
print("Standard Deviation:",std_dev)
```

Standard Deviation: 2.0

### 7. Using Built-in Module

```
In [9]: import statistics

data = [12,15,15,18,22,25]
print("Mean:", statistics.mean(data))
print("Median:", statistics.median(data))
print("Mode:", statistics.mode(data))
print("Variance:", statistics.variance(data))
print("Standard Deviation:", statistics.stdev(data))
```

Mean: 17.833333333333332  
Median: 16.5  
Mode: 15  
Variance: 23.766666666666666  
Standard Deviation: 4.875106836436168

### 8. Using NumPy Library

```
In [10]: import numpy as np
data = np.array([10,20,20,30,40])
print("Mean:", np.mean(data))
print("Median:", np.median(data))
print("Variance (sample):", np.var(data, ddof=1))
print("Standard Deviation :", np.std(data, ddof=1))
print("Range:", np.ptp(data))
```

Mean: 24.0  
Median: 20.0  
Variance (sample): 130.0  
Standard Deviation : 11.40175425099138  
Range: 30

### 9. Using SciPy for Mode

```
In [11]: from scipy import stats

data = [5, 2, 5, 3, 5, 4]
mode_result = stats.mode(data, keepdims=True)
print("Mode:", mode_result.mode[0])
print("Count:", mode_result.count[0])
```

Mode: 5  
Count: 3

### 10. Using Pandas for Complete Summary

In [12]: `import pandas as pd`

```
df = pd.DataFrame({'values': [10, 20, 30, 40, 50, 50]})  
print("Mean:", df['values'].mean())  
print("Median:", df['values'].median())  
print("Mode:", df['values'].mode().tolist())  
print("Variance:", df['values'].var())  
print("Std Dev:", df['values'].std())  
print("Range:", df['values'].max() - df['values'].min())
```

Mean: 33.333333333333336  
Median: 35.0  
Mode: [50]  
Variance: 266.6666666666667  
Std Dev: 16.32993161855452  
Range: 40

In [ ]: